

$$\textcircled{1} \frac{\textcircled{1}}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2} = \frac{Ax + 2A + Bx + B}{(x+1)(x+2)}$$

$$2A + B = 1$$

$$A + B = 0$$

$$A = 1, B = -1$$

$$\left(\frac{1}{x+1} - \frac{1}{x+2} \right)$$

$$\ln|x+1| - \ln|x+2| = \ln\left|\frac{x+1}{x+2}\right| + C$$

$$\textcircled{2} \frac{x^\checkmark}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2} = \frac{Ax - A + B}{(x-1)^2}$$

$$A = B$$

$$A = 1$$

$$\frac{1}{x-1} + \frac{1}{(x-1)^2}$$

$$\rightarrow \ln|x-1| - \frac{1}{(x-1)} + C$$

$$\textcircled{3} \quad \frac{1}{x(x^2+1)} = \frac{\overset{f}{A}}{\underline{x}} + \frac{\overset{f}{Bx+C}}{\underline{x^2+1}} = \frac{Ax^2 + Ax + Bx + C}{x(x^2+1)}$$

$$A = 1$$

$$C = 0$$

$$A + B = 0$$

$$B = \underline{-1}, \quad \left(\frac{1}{x} - \frac{x}{x^2+1} \right)$$

$$\hookrightarrow \ln|x| - \frac{1}{2} \ln(x^2+1) + C$$

$$\textcircled{1} \int \underline{\underline{x \cos x}} dx,$$

$$\int u v dx = u \int v dx - \int (u' v dx) dx$$

$$u = x, \quad v = \cos x$$

$$u' = 1, \quad \int v dx = \sin x$$

$$= x \sin x - \int \sin x dx$$

$$= x \sin x + \cos x \rightarrow \underline{\underline{x \cos x}} + \sin x - \sin x$$

$$\textcircled{2} \int \underline{\underline{\ln x}} dx$$

$$u = \ln x, \quad v = 1$$

$$u' = 1/x, \quad \int v dx = x$$

$$= (\ln x) x - \int \frac{1}{x} x dx$$

$$= x \ln x - x + C$$

$$\textcircled{3} \quad I = \int e^x \sin x \, dx, \quad \begin{array}{l} u = \underline{\sin x}, \quad v = e^x \\ u' = \cos x, \quad \int v \, dx = e^x. \end{array}$$

$$I = (\sin x) e^x - \int \underline{\cos x} \cdot \underline{e^x} \, dx$$

$$= e^x \cdot \sin x - \left[\cos x \cdot e^x + \int \sin x \cdot e^x \, dx \right]$$

$$= e^x (\sin x - \cos x) - I$$

$$\Rightarrow 2I = e^x (\sin x - \cos x)$$

$$I = \underline{\underline{\frac{1}{2} e^x (\sin x - \cos x)}}$$

HW

$$\underline{\underline{\int e^x \cos x \, dx}}$$

$$e^{ix} = \cos x + i \sin x \quad (\text{Euler})$$

$$\operatorname{Re}(e^{ix}) = \cos x$$

$$\operatorname{Im}(e^{ix}) = \sin x$$

$$I = \int \underline{e^x \sin x} \, dx = \underline{\operatorname{Im} \left(\int e^x \cdot e^{ix} \, dx \right)}$$

$$\int e^{x+ix} \, dx = \int e^{x(1+i)} \, dx = \frac{e^{x(1+i)}}{1+i}$$

$$= e^x \cdot \frac{e^{ix}}{1+i} = \frac{e^x}{1+i} (\cos x + i \sin x)$$

$$= \frac{e^x (1-i)}{(1+i)(1-i)} (\cos x + i \sin x)$$

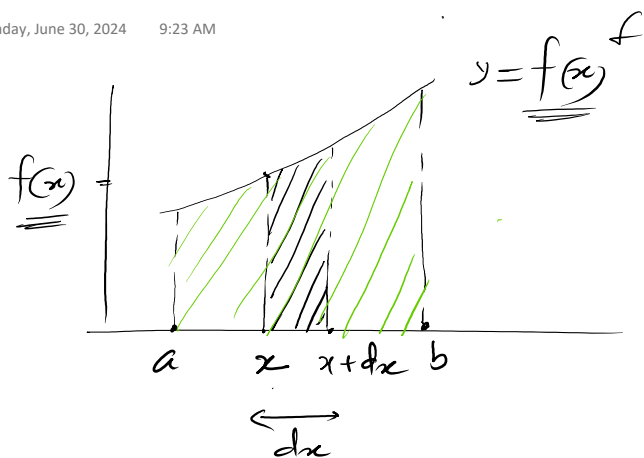
$$= \frac{e^x}{2} (\cos x + i \sin x - i \cos x + \sin x)$$

$$= \underline{\underline{\frac{e^x}{2} ((\cos x + \sin x) + i (\sin x - \cos x))}}$$

$$\cancel{\frac{e^x}{2} (\sin x - \cos x)}$$

$$\int \underline{e^x \cos x} \, dx = \operatorname{Re} \left[\int e^x \cdot e^{ix} \, dx \right]$$

$$\frac{e^x}{2} (\sin x + \cos x)$$



$$\int_a^b \underline{f(x)} dx$$

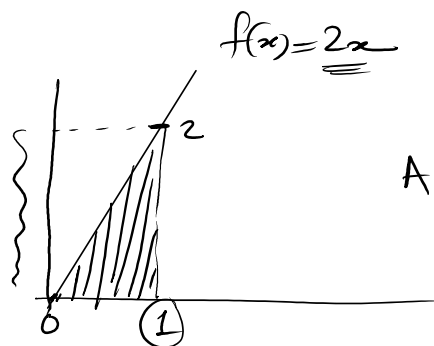
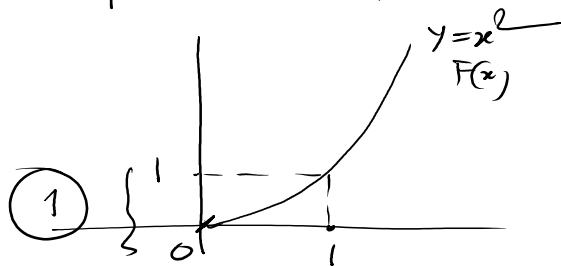
$$\int \underline{f(x)} dx = \underline{F(x)} + C$$

$$\int_a^b \underline{f(x)} dx = \underline{F(b)} - \underline{F(a)}$$

$$m = \frac{dy}{dx} \quad \int m dx = y$$

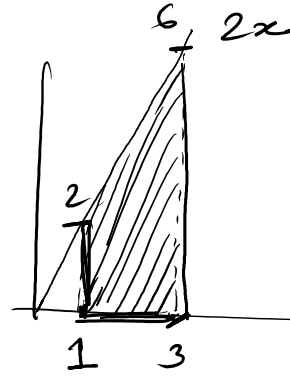
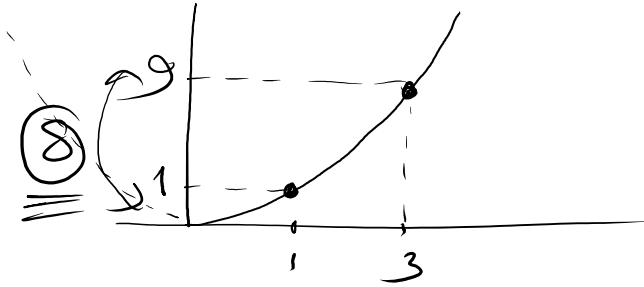
$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int_0^1 (2x) dx = \left[x^2 \right]_0^1 = 1 - 0 = 1$$



$$A = \frac{1}{2} \times 1 \times 2 = 1$$

$$\int_1^3 (2x) dx = \left[x^2 \right]_1^3 = 9 - 1 = 8.$$

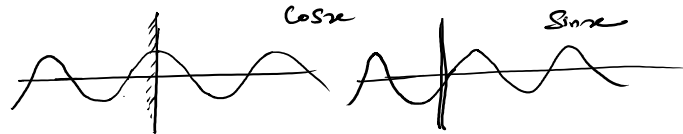


$$A = \frac{1}{2} \times 2 \times (8)$$

$$= 8$$

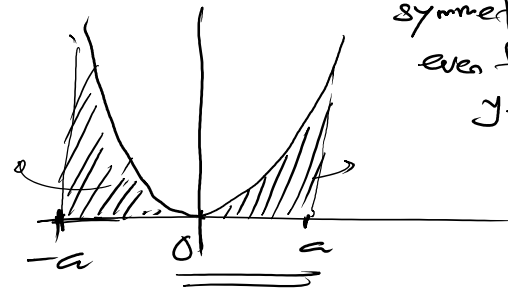
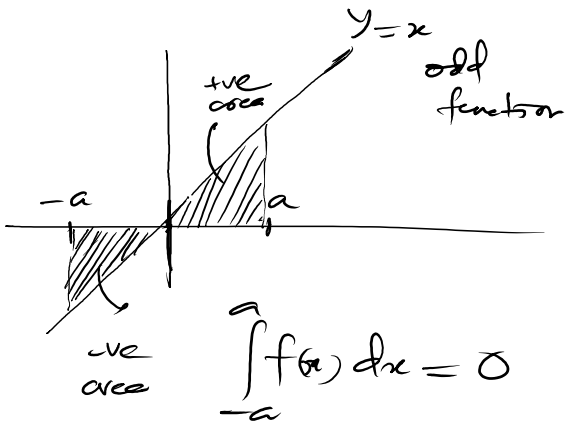
if $f(-x) = f(x) \rightarrow$ even function.

if $f(-x) = -f(x) \rightarrow$ odd function. $\rightarrow$



$$\cos(-x) = \cos x$$

$$\sin(-x) = -\sin x.$$



$$\int_{-a}^a f(x) dx = 2 \times \int_0^a f(x) dx$$

$$\int_{-\pi}^{\pi} x^{\log} dx = 0$$